

铜·铝合金用2刃微小径圆鼻铣刀
Copper · Aluminum Alloy 2-Flute Micro Diameter Corner Radius Endmills

2刃·带R角

2-Flute · Corner Radius

平面

侧切

槽

封底

曲面

R

螺旋

粗加工

半精加工

精加工

切削条件

外径公差

$D \leq 0.1: 0 \sim -0.005$
 $0.15 \leq D \leq 0.8: 0 \sim -0.007$

位置公差

± 0.005

硬度

$0 \sim -0.005$
(mm)

高耐溶着性能。最适合铝合金·铜的加工。tac涂层。
High welding resistance makes it ideal for machining aluminum alloys and copper.tac coating.



- ✦ 针对铜电极加工用的球头铣刀。
- ✦ 采用优化的刀刃形状和tac涂层，可确保长时间进行持续稳定的高品质加工。
- ✦ Ball end milling cutter for copper electrode machining.
- ✦ High quality and stable milling performance with long tool life by optimized design and tac COATING.



产品代码 Code No.	规格型号 Spec Typ.	外径(D) Dia.	角半径(R) Corner Radius	刃长(l) Length of Cut	颈角(r) Neck Taper Angle	柄径(d) Shank Dia.	全长(L) Overall Length	形状 Type	定价 Retail Price
FX3SR01001	0.1R0.01x0.2L	0.1	R0.01	0.2	10+20°	4	50	A	
FX3SR01002	0.1R0.02x0.2L	0.1	R0.02	0.2	10+20°	4	50	A	
FX3SR015002	0.15R0.02x0.3L	0.15	R0.02	0.3	10+20°	4	50	A	
FX3SR015003	0.15R0.05x0.3L	0.15	R0.05	0.3	10+20°	4	50	A	
FX3SR02002	0.2R0.02x0.4L	0.2	R0.02	0.4	10+20°	4	50	A	
FX3SR02005	0.2R0.05x0.4L	0.2	R0.05	0.4	10+20°	4	50	A	
FX3SR03005	0.3R0.05x0.6L	0.3	R0.05	0.6	10+20°	4	50	A	
FX3SR0301	0.3R0.1x0.6L	0.3	R0.1	0.6	10+20°	4	50	A	
FX3SR04005	0.4R0.05x0.8L	0.4	R0.05	0.8	10+20°	4	50	A	
FX3SR0401	0.4R0.1x0.8L	0.4	R0.1	0.8	10+20°	4	50	A	
FX3SR05005	0.5R0.05x1.0L	0.5	R0.05	1	10+20°	4	50	A	
FX3SR0501	0.5R0.1x1.0L	0.5	R0.1	1	10+20°	4	50	A	
FX3SR06005	0.6R0.05x1.2L	0.6	R0.05	1.2	10+20°	4	50	A	
FX3SR0601	0.6R0.1x1.2L	0.6	R0.1	1.2	10+20°	4	50	A	
FX3SR07005	0.7R0.05x1.4L	0.7	R0.05	1.4	10+20°	4	50	A	
FX3SR0701	0.7R0.1x1.4L	0.7	R0.1	1.4	10+20°	4	50	A	
FX3SR08005	0.8R0.05x1.6L	0.8	R0.05	1.6	10+20°	4	50	A	
FX3SR0801	0.8R0.1x1.6L	0.8	R0.1	1.6	10+20°	4	50	A	
FX3SR0802	0.8R0.2x1.6L	0.8	R0.2	1.6	10+20°	4	50	A	

切削参数参考表
Recommended Milling Conditions

加工材料 Work Material		铜合金 Copper			
外径 Dia.	角半径(R) Corner Radius	主轴转速 Spindle Speed	进给速度 Feed	切深量 Depth of Cut	
		min-1	mm/min	ap mm	ae mm
0.1	0.01	40,000	200	0.005	0.005
	0.02	40,000	200	0.005	0.005
0.15	0.02	40,000	300	0.008	0.008
	0.05	40,000	300	0.008	0.008
0.2	0.02	40,000	400	0.01	0.1
	0.05	40,000	400	0.03	0.1
0.3	0.05	40,000	480	0.03	0.15
	0.1	40,000	480	0.06	0.15
0.4	0.05	40,000	640	0.03	0.2
	0.1	40,000	640	0.06	0.2
0.5	0.05	40,000	800	0.03	0.25
	0.1	40,000	800	0.06	0.25
0.6	0.05	30,000	1,000	0.03	0.3
	0.1	30,000	1,000	0.06	0.3
0.7	0.05	30,000	1,500	0.03	0.35
	0.1	30,000	1,500	0.06	0.35
0.8	0.05	30,000	1,600	0.03	0.4
	0.1	30,000	1,600	0.06	0.4
0.8	0.2	30,000	1,600	0.06	0.4
备注 Notes		※本切削参数仅供参考。请根据实际的加工形状和所使用的机床等调整切削参数。 ※切深量的ap表示轴向切深量，ae表示径向切深量。 ※轴向进刀建议采用螺旋进刀及倾斜进刀方式。 ※沟槽切削时建议参考切削参数表，进给速度设定为60%以下，并采用来回切削加工方式。 ※发生振刀时，请以相同的比率降低主轴转速和进给速度。此外，主轴转速过低时，也以相同的比率降低。 ※建议使用油冷冷却方式。 ※ The cutting parameters are for reference only. Please adjust the cutting parameters according to the actual machining shape and the machine tool used. ※ The ap of the cutting depth indicates the axial cutting depth, and the ae indicates the radial cutting depth. ※ Screw feed and inclined feed are recommended for axial feed. ※ It is recommended to refer to the cutting parameter table when grooving cutting, the feed speed is set to less than 60%, and the cutting mode is adopted. ※ When vibrating knife occurs, please reduce the spindle speed and feed speed in the same ratio. In addition, when the spindle speed is too low, it is also reduced at the same rate. ※ Oil cooling is recommended.			